

DETERMINISTIC MEMORY FOR AI AGENTS

Recall. Recognize. Adapt.

An embedded, embedding-free memory your agent can trust — what do I know, have I seen this before, what matters now — answered from one SQLite file. No vector database. No GPU. No LLM on the recall path.

- \$0 recall

Local-first

Byte-reproducible

Federation-ready

SIX KINDS OF MEMORY · ONE BRAIN

The memory your agent gets

Each mode is deterministic and embedding-free, so every one is free to query — no model call, no encoding step, no per-query bill.

Recall	“What do I know about X?” SQLite FTS5 + BM25, porter-stemmed, ranked by signal-score & recency.	\$0 · deterministic
Recognition	“Have I seen this before — and is it new?” Landmark fingerprinting → a familiarity / novelty verdict carrying the exact matched features.	\$0 · deterministic
Relational	“How does X relate to Y?” Typed knowledge graph with ontology validation and 2-hop traversal.	\$0 · deterministic
Episodic	“What happened around then?” Session episodes with time-anchored recency ranking and decay.	\$0 · deterministic
Adaptive	“What matters <i>now</i> ?” Use-driven reinforcement, disuse decay, and lift-based anticipation — a memory that learns from use.	\$0 · deterministic
Federated	“What do we collectively know?” Read-time fan-out across brains — provenance-ranked, visibility-scoped, poisoning-resistant.	\$0 · deterministic

On **LongMemEval-S** it handles every memory-question type — single-session (user, assistant, preference), multi-session, temporal-reasoning, and knowledge-update — at **98.6% session-recall**, embedding-free.

THE ECONOMICS

Memory that’s free by construction

Not “cheap” — structurally free. Recall and recognition make zero model calls, so you can *assert* the memory layer adds no per-query cost, not just hope it stays small.

	SPECTRAL	EMBEDDING / VECTOR-DB MEMORY
recall query	\$0 — no model, no API call	an embedding call per query (often + an LLM)
ingest	\$0 — deterministic classify + index	an embedding call per item
infrastructure	one embedded SQLite file	a vector-DB service / server + index
hardware	CPU only — no GPU, no weights	GPU or a hosted embedding endpoint
determinism	byte-reproducible	drifts with the model version
ZERO-LLM	recognition_token_cost == 0 is structural — the memory layer’s LLM cost is provably nil.	
OPT-IN	The only optional spend is a pre-retrieval query-expansion call (Claude Haiku, ≈ \$0.25 / 1k queries) that lifts paraphrase accuracy. Off unless you enable it.	
LATENCY	Sub-millisecond recall on 1k memories; ~15–21 ms on LongMemEval-scale haystacks — the adaptive write-back is batched, and optionally moved off the recall path entirely.	
STORAGE	A single SQLite file you own — copy it, back it up, delete it. Local-first by construction.	

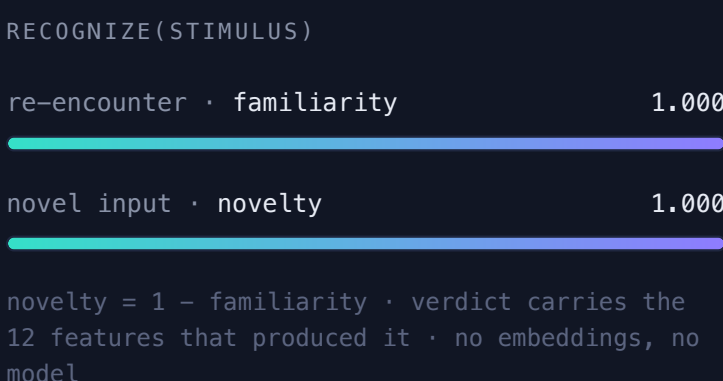
WHERE THE NAME COMES FROM

Recognition: “have I seen this?”

Most memory only *recalls*. Spectral also **recognizes** — scoring a new stimulus as familiar or novel, deterministically, with the exact features behind the verdict.

Its landmarks are a stimulus’s statistically salient features — the text analog of **spectral peaks above the noise floor**. That’s the name.

- Landmark fingerprinting — IDF-salient features (Shazam-style)
- Winnowed k-grams — the MOSS verbatim-run guarantee
- REM + MINERVA-2 — cognitive-psych familiarity scoring



MEASURED, NOT ASSERTED

Proven robust, stable, and free

98.6% session-recall across all six memory types (LongMemEval-S)	\$0 structural LLM cost on the recall path	−48% recall latency after batching the adaptive write-back	640 concurrent read/write ops, 8 threads — no deadlock, no corruption
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Every figure is backed by a test that fails without it. In-sample retrieval; held-out expected lower. Full method & limitations in the repo.